

Question 1: Which members of the Circle class are encapsulated?

The private data members, such as the radius, are encapsulated because they are hidden and can only be accessed using public methods.

Question 2: What name must the constructor of a class have?

The constructor must have the same name as the class and it does not have a return type.

Question 3: Explain the difference between private and public.

Private means the variable or method can only be used inside the class.

Public means the variable or method can be used by any class.

In short, private hides information and public allows access.

Question 4: Consider the example where a Circle object is created with a radius of 2, then the radius is changed to 5 directly. Is this valid or invalid? Explain.

This is invalid if the radius is private. You cannot change a private variable directly. You must use a setter method instead.

Question 5: Using the Roo class shown, answer:

(a) What is the name of the class?

The name of the class is Roo.

(b) What is the name of the data member?

The data member is x.

(c) List the accessor method.

The accessor method is get X.

(d) List the modifier method.

The modifier method is set X.

(e) List the helper method.

The helper method is factor.

(f) What is the name of the constructor?

The constructor is named Roo.

(g) How many method members are there?

There are five total method members, which include four regular methods and one constructor.

Question 6: What is the difference between a class and an object?

A class is a blueprint or template.

An object is a real example made from the class.

Question 9: Given the data members in Moo which are y as a double, x as a static integer, and z as a static final value:

(a) Which data member is a constant?

The constant is z.

(b) Which data members are variables?

The variables are y and x.

(c) Which data member is an instance member?

The instance member is y.

(d) Which data members are class members?

The class members are x and z.

Question 11: Compare overriding and overloading.

Overriding happens when a subclass replaces a method from its parent class using the same method name and same parameters. It is decided at run time.

Overloading happens when a class has more than one method with the same name but different parameters. It is decided at compile time.